Exercise 5

BIOENG-310: Neuroscience Foundations for Engineers

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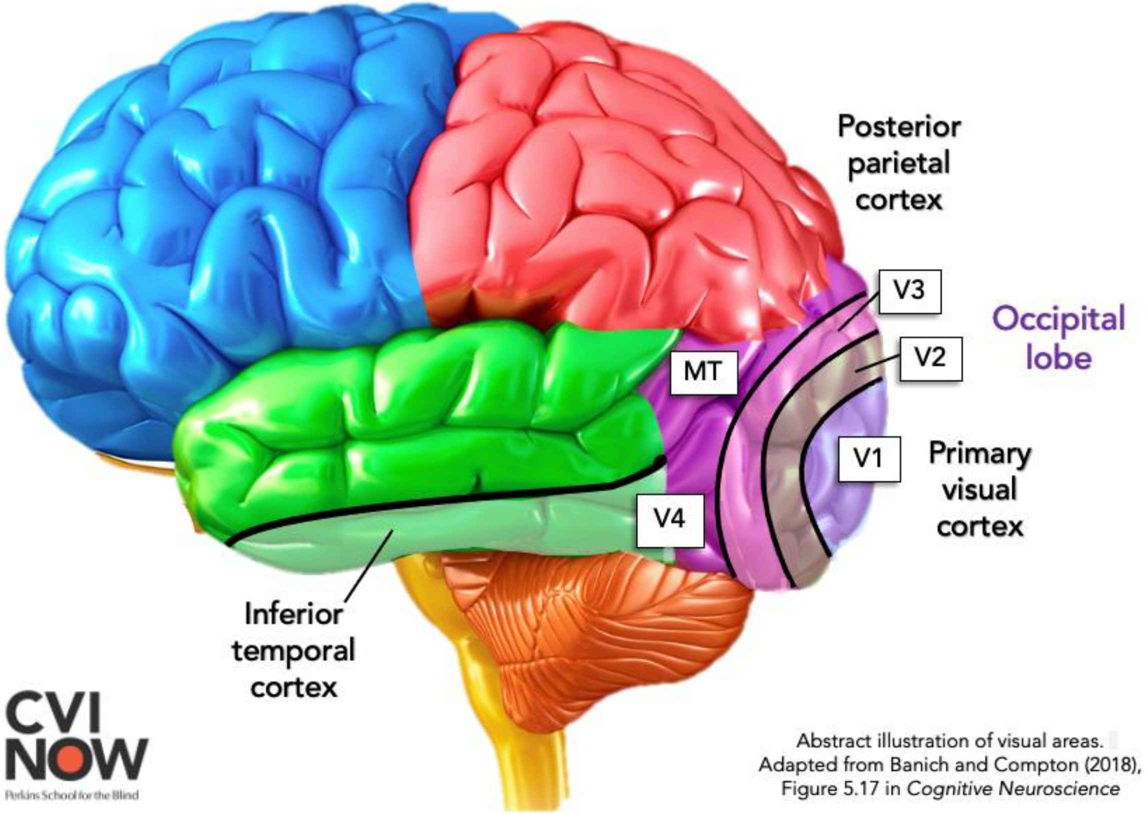
Context of the Exercise

- We will use a dataset from Freeman & Ziemba, 2013, a study on V1 and V2 responses to visual textures in macaque monkeys.

- **Objective:** Learn how to manipulate and interpret neural response data.

Overview of visual cortex V1 and V2



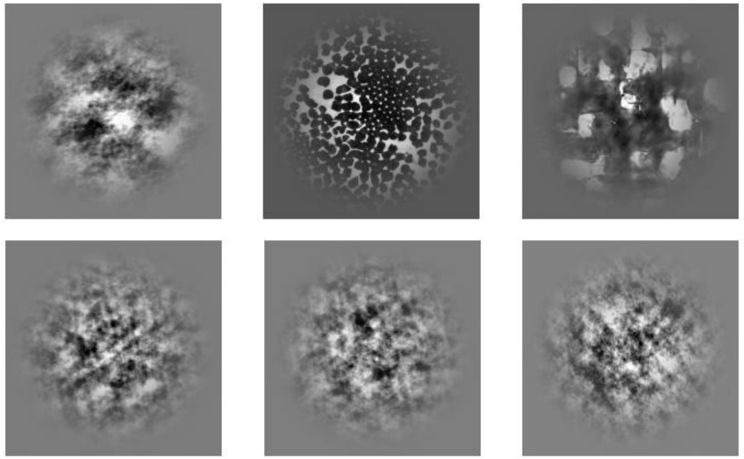
Abstract illustration of visual areas.
Adapted from Banich and Compton (2018),
Figure 5.17 in *Cognitive Neuroscience*

- **V1** → Detects basic visual features: edges, orientation, contrast. **Simple feature detection**
- **V2** → Builds on V1 information to detect **textures and complex patterns**.

Experimental Methodology

Subjects: Macaque monkeys.

Stimuli: Synthetic images were generated to match natural textures or contain randomized noise.

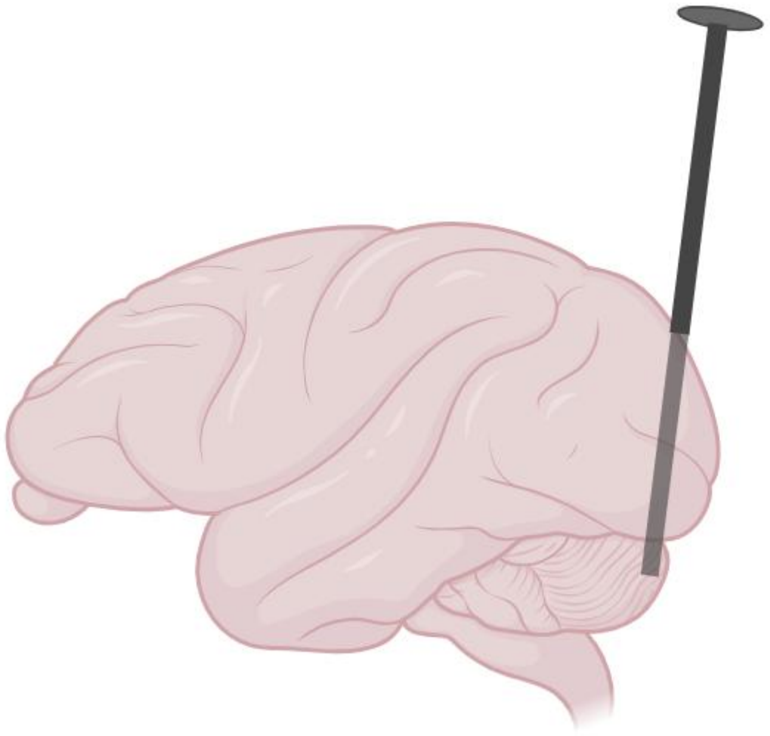


Experimental Methodology

Electrophysiological recordings:

Several microelectrodes implanted in the V1 and V2 cortex.

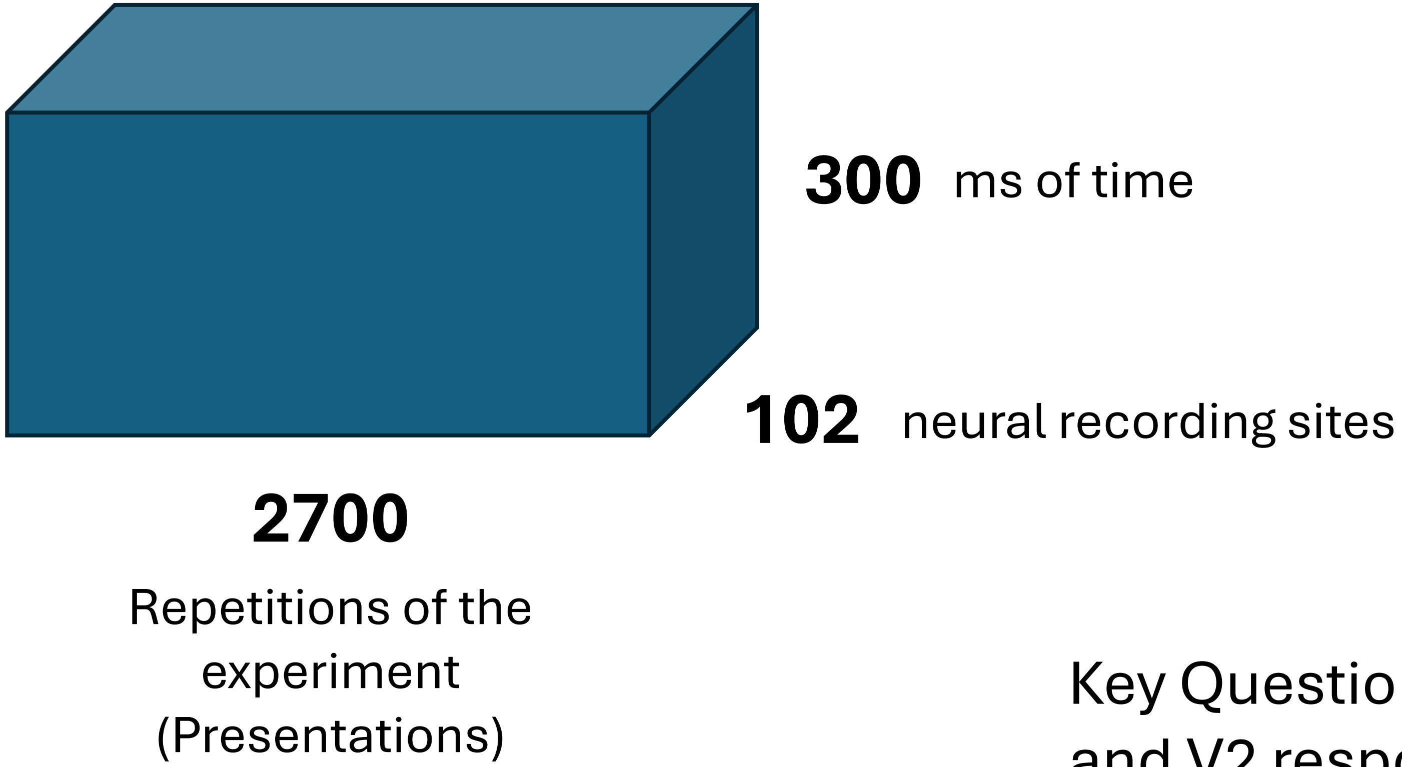
These electrodes are used to record **Local Field Potentials (LFP)**.



102 recording sites!

The Dataset

Data by Freeman & Ziemba et al. 2013, packaged into the Brain-Score platform.

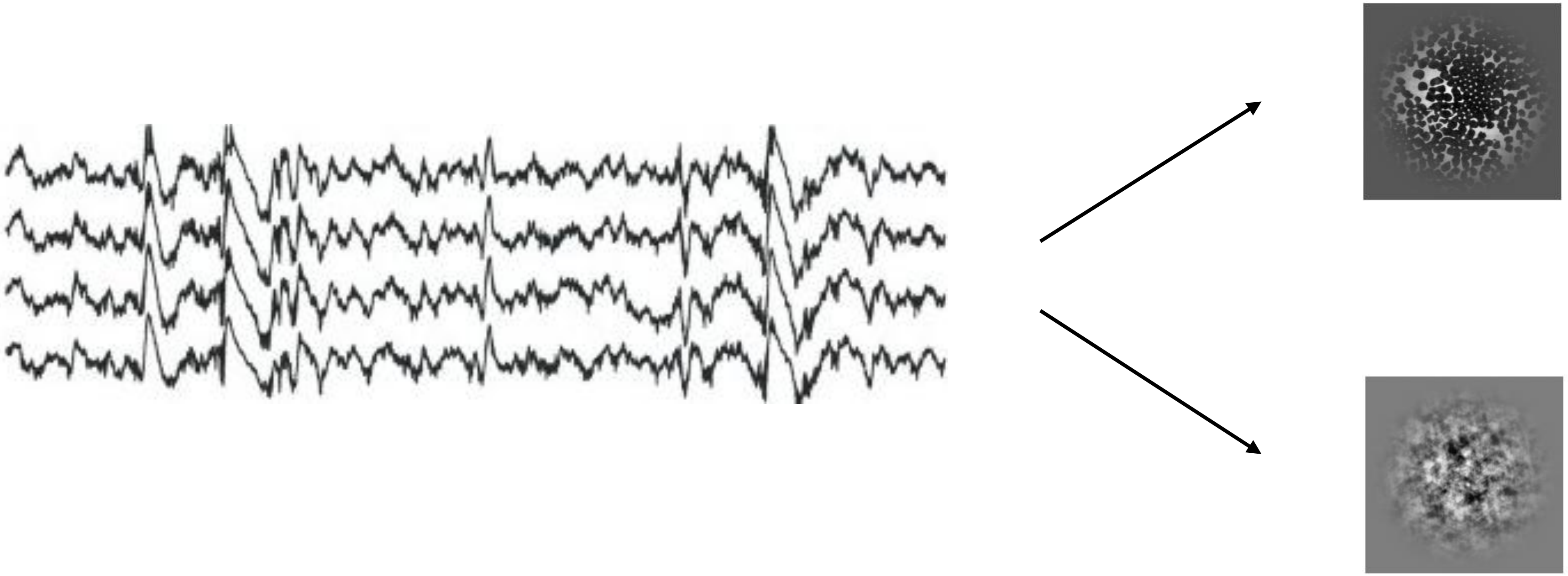


Key Question: How do V1 and V2 respond differently to naturalistic textures?

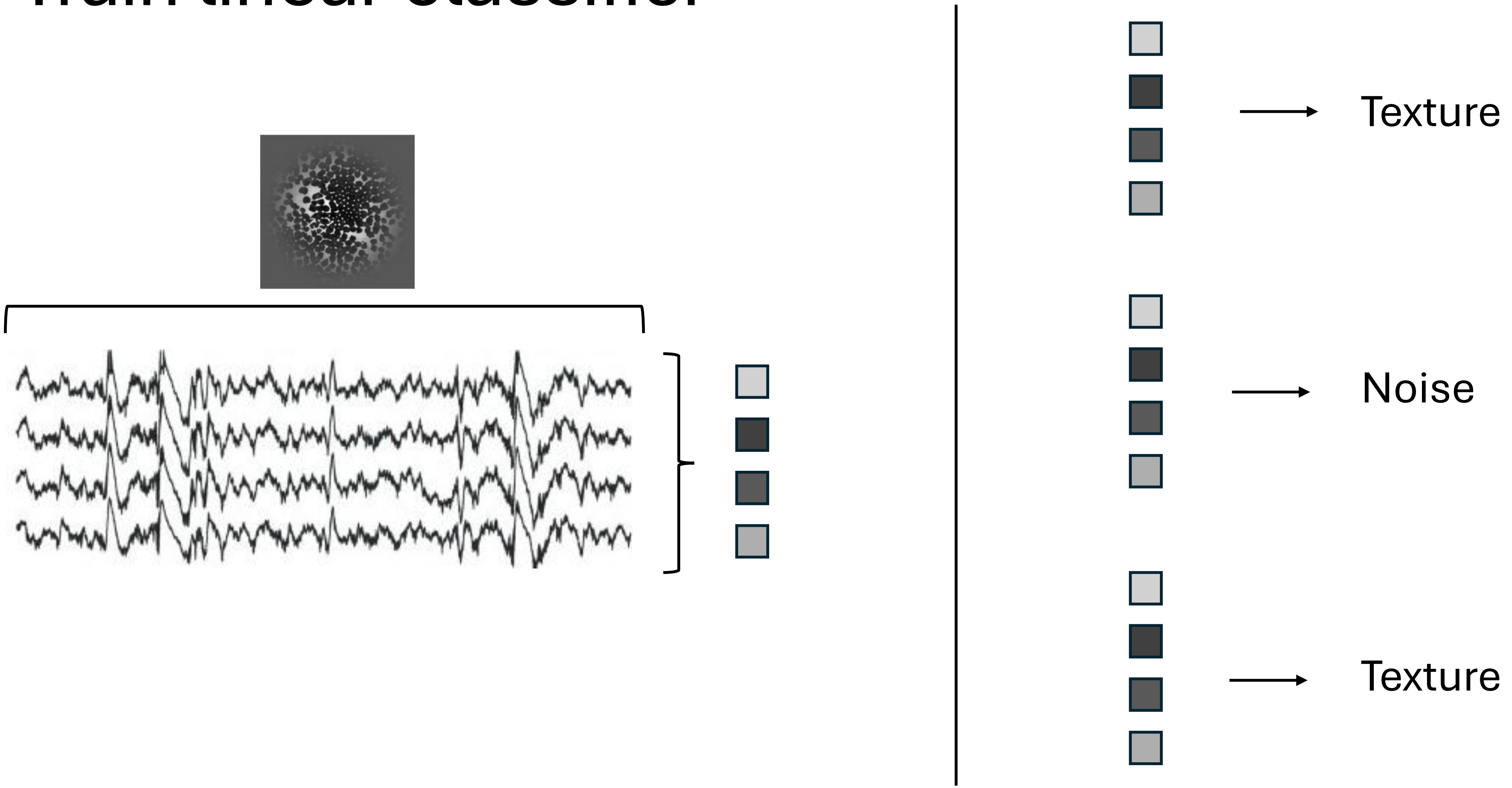
Objectives

- Download the dataset
- Access and visualize the different stimulus
- Access and visualize the neural data
- **Apply Linear readout / linear probing to sort the data**

Retrieve image type from LFP



Train linear classifier



Questions?